

Course Code : ECT 456-3

KQLR/RS – 24/ 1316

**Eighth Semester B. Tech. (Electronics and Communication
Engineering) Examination**

(Elective – VI)

ANTENNA THEORY

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
- (2) Assume suitable data wherever necessary.
- (3) All questions carry marks as indicated.

1.
 - (a) Explain the current distribution on a two wire antenna. 5(CO1)
 - (b) Calculate the maximum effective aperture of a microwave antenna which has directivity of 900. Also calculate efficiency of the antenna if its radiation resistance is $80\ \Omega$ and load resistance is $20\ \Omega$. 5(CO4)
2.
 - (a) Explain the Huygens' principle for field equivalence with suitable example. 5(CO1,2)
 - (b) Explain with diagrams constructional details of various types of Horn antennas. 5(CO1,2)
3.
 - (a) Differentiate between Log periodic and Yagi Uda antennas. 5(CO2)
 - (b) Discuss various feed mechanisms for micro strip patch antenna in order to reduce spurious radiations. Explain with diagram. 5(CO2)
4.
 - (a) Compare Broadside array antenna with End fire array antenna system in detail. 5(CO5)
 - (b) For two element array consisting of identical radiators carrying equal currents in phase, obtain positions of maxima and minima of the radiation pattern if the distance of separation $d = \lambda$. 5(CO4)

KQLR/RS-24 / 1316

Contd.

5. (a) Compare with diagrams the co-channel interference of Omnidirectional base station, Sectorized base station and Smart antenna systems. 5(CO2,3)
- (b) Explain the benefits of Smart antennas over conventional antennas. 5(CO2,3)
6. (a) Explain the architecture of smart antenna system in detail. 5(CO3)
- (b) Design with block diagram a suitable antenna system for the following application :
A 5 G Mobile communication system to be set up within 2 Km area from a base station. 5(CO3)

